

BHOOMIKA SHARMA

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SUMMARY

Data & Machine Learning Engineer specializing in predictive modeling, computer vision, and scalable AI architectures. Proficient in leveraging Python and SQL to build end-to-end data pipelines and deploy high-performance machine learning models. Focused on transforming complex datasets into production-grade solutions that drive actionable business impact.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science Engineering	2023–2027
Dr. A.P.J. Abdul Kalam Technical University, Lucknow — GL Bajaj Group of Institutions, Mathura	
Senior Secondary Education – Central Board of Secondary Education (80%)	2022–2023
Secondary Education – Central Board of Secondary Education (96%)	2020–2021

TECHNICAL SKILLS

Programming:	Python, SQL
ML & Computer Vision:	PyTorch, TensorFlow, TensorRT, Scikit-learn, OpenCV, YOLO, Mediapipe
Tools:	Git, Docker
Concepts:	Data Structures & Algorithms (DSA), Machine Learning, Deep Learning, Computer Vision, Object Detection, Image Segmentation, Vision-Language Models (VLM), Edge AI, Real-time Inference

PROJECTS

Samvita (NabhaDrishti) — Geospatial Intelligence System

- Architected a YOLOv11X segmentation pipeline for land feature extraction from multi-GB 4K UAV imagery, achieving 93.5% mAP and reducing manual GIS workflows by 85%.
- Engineered a multithreaded OpenCV pipeline processing 500+ GB/day and optimized inference via TensorRT FP16, decreasing latency to <40ms/tile while reducing GPU utilization by 40%.

Multimodal Visual Question Answering (VQA) System

- Developed a multimodal VQA system integrating BLIP-2 and RoBERTa, achieving 78.4% accuracy after fine-tuning on 1.2M VQA_{v2} image-text pairs.
- Built a Ray-based distributed preprocessing pipeline handling 500GB daily and slashed inference latency to 22ms/query via TensorRT INT8 quantization.

NVDamis — Real-Time Driver Monitoring System

- Designed a real-time driver monitoring pipeline leveraging YOLOv11X, achieving 96.5% mAP for micro-sleep detection across a 200GB dataset.
- Accelerated edge inference to 12ms latency at 60 FPS via TensorRT INT8 optimization and implemented a multi-threaded OpenCV architecture to guarantee zero frame drops.

ACHIEVEMENTS

- Led a team in the Q2-Finxthon national hackathon, developing and presenting a fintech solution under competitive conditions.
- Presented innovative technology solutions at Innovative Bharat 2.0 & 3.0, collaborating across teams in inter-institutional events.
- Coordinated large-scale college events including the IEEE Conference 2025, managing logistics, operations, and team execution.
- Completed Cisco certifications: Exploring Networking with Packet Tracer, Introduction to Cybersecurity, and Junior Cybersecurity Analyst Path.